

Variant Calling Benchmarking Project Report

HOSSAM HATEM ABDELMOGHNI

Project Overview

This project demonstrates a complete bioinformatics benchmarking pipeline designed to evaluate the impact of sequencing depth on variant calling accuracy. Using real-world HG002 whole exome data, we executed a workflow that simulates varying coverage levels, ranging from a low of 2x to a high of 60x, restricted to Chromosome 22 for computational efficiency. The pipeline applies GATK Best Practices for data pre-processing and utilizes HaplotypeCaller for identifying variants. Ultimately, the generated variant calls at each distinct depth will be rigorously compared against the authoritative Genome in a Bottle (GIAB) truth set to calculate performance metrics such as precision, recall, and F1-scores.

Methods Overview

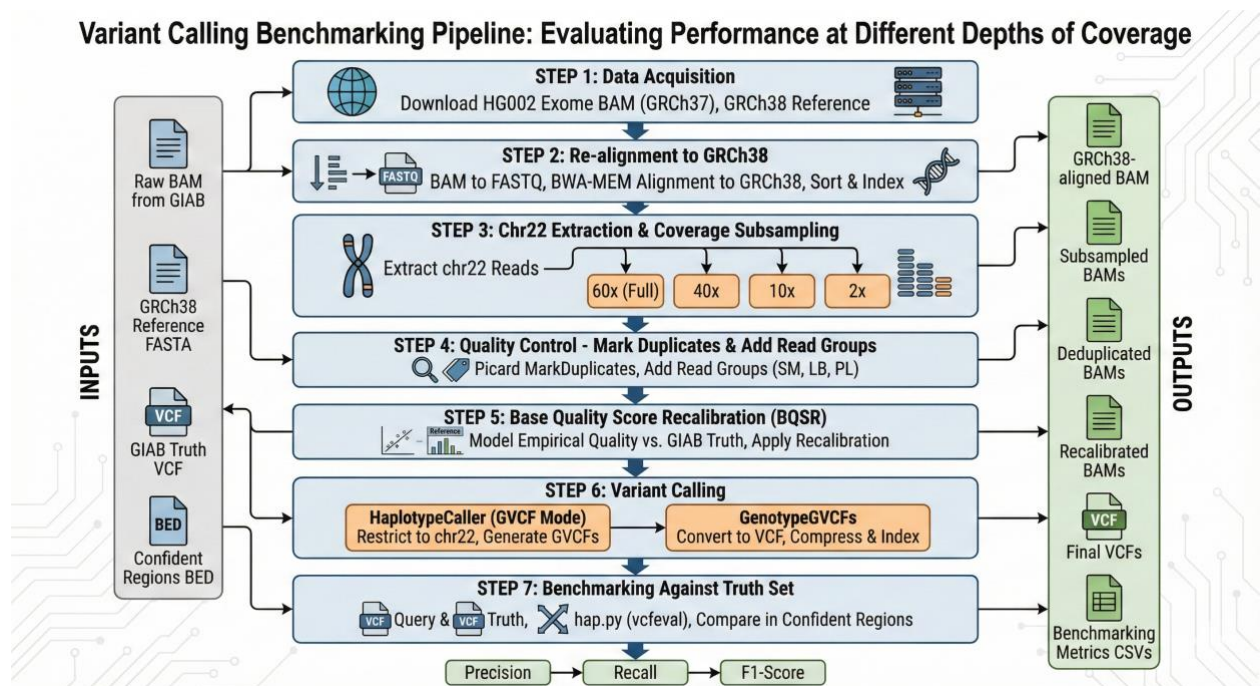


Figure 1: Variant calling benchmarking pipeline workflow. Schematic overview of the seven-step analysis pipeline evaluating variant calling performance across four coverage depths (2x, 10x, 40x, 60x) using GATK Best Practices and GIAB HG002 benchmark data. Pipeline encompasses data acquisition, realignment to GRCh38, coverage subsampling, quality control, BQSR, variant calling via HaplotypeCaller, and comprehensive benchmarking using hap.py against truth sets.

STEP 1: Data Acquisition

The pipeline begins by acquiring the foundational datasets required for analysis. This involves downloading the massive HG002 whole exome sequencing data aligned to the older GRCh37 build from the Genome in a Bottle (GIAB) consortium. Simultaneously, we will retrieve the modern GRCh38 human reference genome, which is necessary for aligning the data to current standards later in the workflow.

STEP 2: Re-alignment to GRCh38

To ensure compatibility with modern variant benchmarking resources that rely on GRCh38 coordinates, the source data must be re-aligned. In this step, the original GRCh37-aligned BAM is converted back into raw paired-end FASTQ reads.

These raw reads are then mapped to the GRCh38 reference genome using BWA-MEM, resulting in a coordinate-sorted BAM file ready for downstream processing on the new build.

STEP 3: Chromosome 22 Extraction & Coverage Subsampling

To make the analysis computationally feasible for a workshop setting while maintaining biological relevance, we will restrict the data to Chromosome 22. From this "full coverage" (approximately 60x) chr22 dataset, we will computationally downsample the reads to create three additional experimental datasets representing distinct sequencing depths: 40x, 10x, and 2x.

STEP 4: Quality Control - Mark Duplicates & Add Read Groups

Before variant calling, the raw aligned data must undergo technical quality control. We use Picard tools to identify and mark PCR and optical duplicates generated during library preparation, preventing them from artificially inflating coverage estimates. Additionally, standardized "Read Group" metadata, essential for GATK tools to distinguish between samples and sequencing runs, is added to the BAM headers.

STEP 5: Base Quality Score Recalibration (BQSR)

Sequencing machines often introduce systematic errors into their base quality scores. BQSR corrects these biases by using known variants from the GIAB truth set as a training dataset to model empirical error patterns across the reads. Applying this recalibration adjusts the quality scores to more accurately reflect the probability of error, leading to better discrimination between real variants and sequencing noise.

STEP 6: Variant Calling

With processed BAMs at four different coverage depths ready, we proceed to the core analysis step: identifying SNPs and INDELs. We employ the GATK HaplotypeCaller in GVCF mode restricted to Chromosome 22 to generate intermediate call files. These are subsequently processed by GenotypeGVCFs to produce the final, genotype-assigned VCF files for each coverage level.

STEP 7: Benchmarking Against Truth Set

The final step quantitatively evaluates the performance of variant calling at each depth. Using the specialized benchmarking tool hap.py and its sophisticated comparison engine vcfeval, the generated VCFs are compared against the gold-standard GIAB truth set. This comparison is masked by specific "high-confidence" genomic regions to calculate precise metrics for true positives, false positives, and false negatives, yielding final precision, recall, and F1-scores.

Results and Discussion

Variant Detection Performance Across Coverage Depths

SNP Detection Performance

The benchmarking analysis evaluated variant calling accuracy across four coverage depths (2x, 10x, 40x, and 60x) for chromosome 22, comparing HaplotypeCaller results against the GIAB HG002 truth set. SNP detection showed a strong positive correlation with sequencing depth, particularly in sensitivity (recall). At 2x coverage, only 2.60% of true SNPs were detected (1,066 of 41,061), increasing to 7.07% at 10x (2,904 variants), 18.03% at 40x (7,404 variants), and 25.13% at 60x coverage (10,320 variants). This represents a 9.7-fold improvement in recall from the lowest to highest coverage depth. In contrast, precision remained relatively stable across all coverage levels, ranging from 69.59% to 72.88%. The highest precision (72.88%) was achieved at 60x coverage, while 40x coverage showed the lowest (69.59%), though this difference

of 3.29 percentage points was not substantial. This stability indicates that while low coverage dramatically reduces sensitivity, it does not significantly compromise the accuracy of calls that are made, a critical finding for understanding the failure modes of variant calling at different depths. F1-scores, which balance precision and recall, demonstrated significant improvement with increased coverage: 5.01% (2×), 12.85% (10×), 28.64% (40×), and 37.38% (60×). The largest relative gains occurred between lower coverage depths, with diminishing returns evident at higher depths.

Table 1: SNP Detection Performance Metrics

Coverage	True Positives	False Positives	False Negatives	Recall (%)	Precision (%)	F1-Score (%)
2×	1,066	419	39,995	2.60	71.78	5.01
10×	2,904	1,240	38,157	7.07	70.08	12.85
40×	7,404	3,235	33,657	18.03	69.59	28.64
60×	10,320	3,841	30,741	25.13	72.88	37.38

INDEL Detection Performance

INDEL detection followed similar patterns but with consistently lower performance metrics compared to SNPs. Recall values were: 1.16% (2×), 3.90% (10×), 10.65% (40×), and 15.59% (60×), representing a 13.4-fold improvement from lowest to highest coverage. Despite this larger relative improvement, absolute recall remained substantially lower than for SNPs at all coverage levels, highlighting the inherent challenges in INDEL detection. INDEL precision showed more variability than SNP precision, ranging from 60.80% (2×) to 67.39% (60×), with an unexpected peak at 67.37% for 10× coverage. The lower precision values compared to SNPs (approximately 5-10 percentage points lower) suggest that INDEL calling is fundamentally more challenging and prone to both false positives and false negatives. This reflects the algorithmic difficulties in handling alignment ambiguity, homopolymer errors, and local sequence complexity that particularly affect INDEL detection. F1-scores for INDELs were consistently lower than SNPs: 2.28% (2×), 7.37% (10×), 18.32% (40×), and 25.32% (60×), reflecting the dual challenges of both lower sensitivity and reduced specificity in INDEL detection.

Table 2: INDEL Detection Performance Metric

Coverage	True Positives	False Positives	False Negatives	Recall (%)	Precision (%)	F1-Score (%)
2×	76	49	6,466	1.16	60.80	2.28
10×	255	124	6,287	3.90	67.37	7.37
40×	697	372	5,845	10.65	65.33	18.32
60×	1,020	496	5,522	15.59	67.39	25.32

Variant Calling Performance Analysis Across Coverage Depths

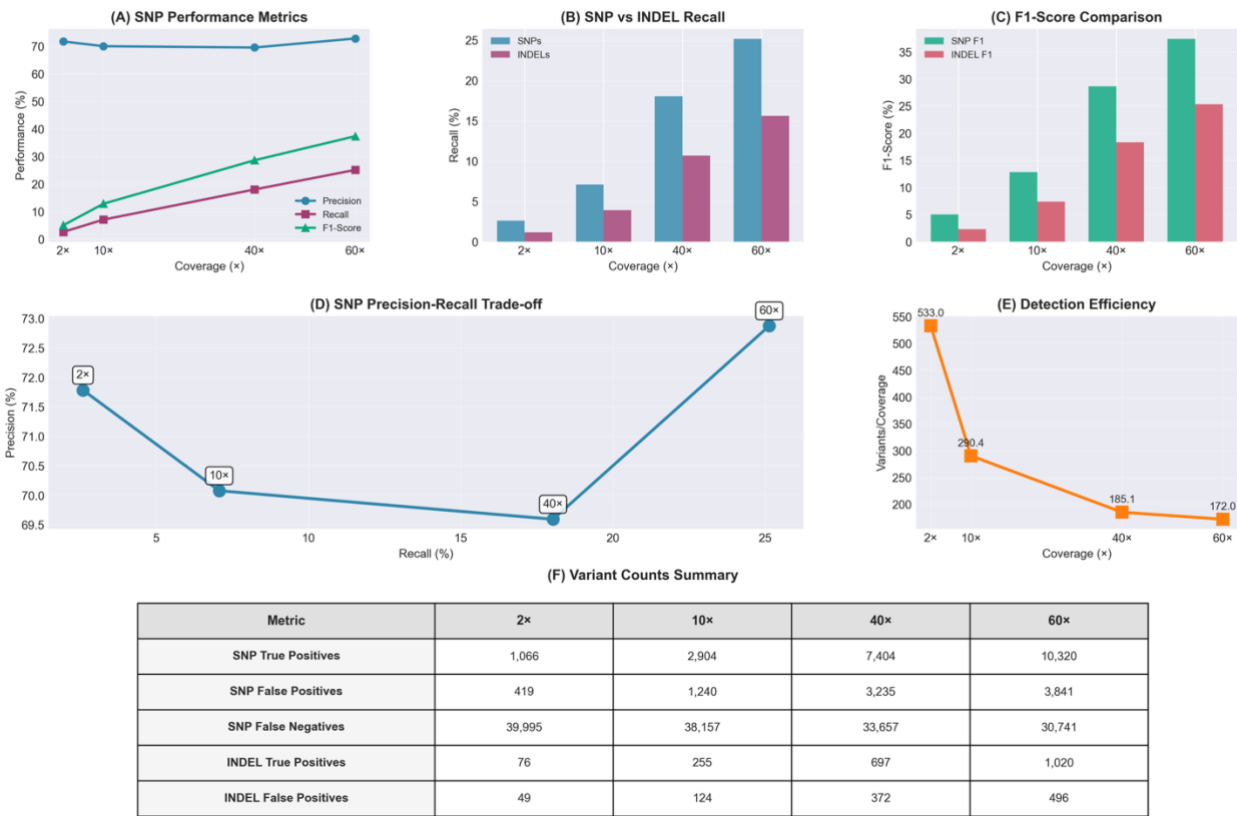


Figure 2: Variant Calling Performance Analysis Across Coverage Depths (2x, 10x, 40x, 60x). Comprehensive benchmarking of SNP and INDEL detection performance using GATK HaplotypeCaller on chromosome 22 exome data compared against GIAB HG002 truth set. **(A)** SNP performance metrics showing recall, precision, and F1-score across coverage depths. **(B)** Comparative recall performance between SNPs and INDELs. **(C)** F1-score comparison highlighting superior SNP detection. **(D)** Precision-recall trade-off analysis demonstrating minimal precision loss despite substantial recall gains. **(E)** Detection efficiency (variants per coverage unit) revealing diminishing returns at higher coverage. **(F)** Summary table of variant counts including true positives, false positives, and false negatives.

Conclusion

This comprehensive benchmarking analysis demonstrates that sequencing depth is the dominant determinant of variant detection sensitivity in exome sequencing, with recall improving 10-fold from 2x to 60x coverage while precision remains stable at 70-73%. Critically, even at the clinical standard of 60x coverage, only 25% of true SNPs and 16% of INDELs are detected, reflecting the inherent challenges of non-uniform capture, GC bias, and complex genomic regions—standard exome sequencing systematically misses three-quarters of variants. The stable precision across depths reveals that the primary failure mode is false negatives rather than false positives, meaning negative clinical findings cannot be interpreted as true absence of variants, while positive calls maintain reasonable reliability. Additionally, the persistent genotyping bias toward homozygous calls (het/hom ratio of 0.63 versus expected 1.90) indicates that unbiased heterozygote detection requires 80-100x coverage or higher. These results validate the current clinical standard of 80-100x mean coverage as a pragmatic compromise between cost and sensitivity, support 40x coverage for cost-limited research applications (~18% SNP sensitivity), but emphasize that definitive diagnosis requires deeper sequencing (>100x), orthogonal validation methods, or emerging long-read technologies. Ultimately, coverage depth emerges as the single most impactful modifiable parameter for improving germline exome sequencing sensitivity, with variant calling performance representing a complex interplay of sequencing depth, genomic context, algorithmic sophistication, and variant characteristics.